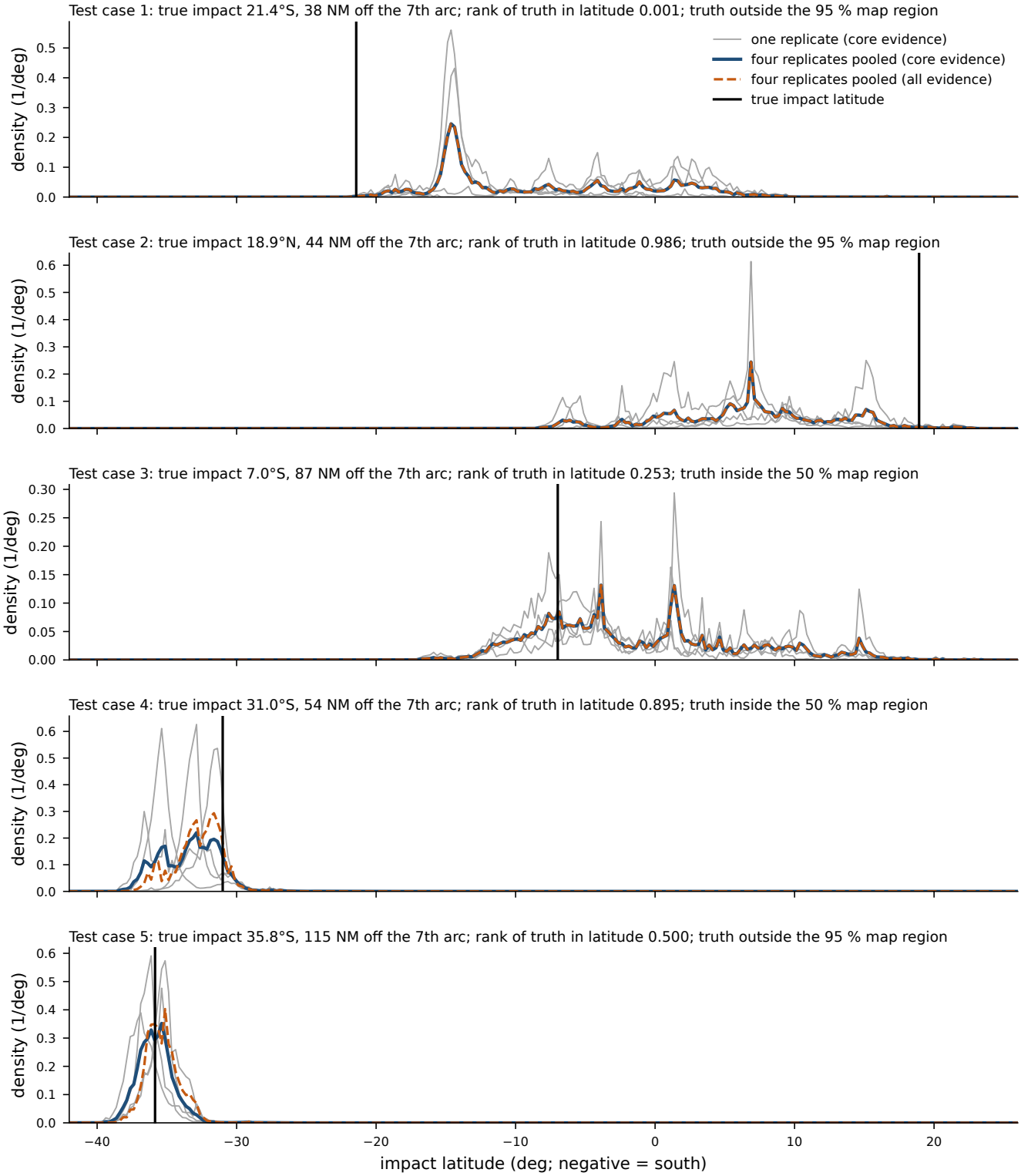
Posterior impact latitude against the true impact for five synthetic test cases (reduced sample size)



Note (simplified English): each panel is one test flight that we made from the model, with known impact point (black line). We ran the full chain on its synthetic data with 4 replicates of 0.7 million particles. Grey lines show each replicate, so their spread shows the Monte Carlo noise. In test cases 1 and 2 the posterior does not contain the truth: the sampler did not find it at this sample size. Map region = highest-density region on an equal-area grid. The 00:19 data are held out. Physics of the end of flight is provisional. The drift and search terms are reduced versions (see note).

Technical: truths drawn by sampling-importance-resampling from the fuel-only engine prior (run J waypoints chain, single-particle replicates; data event BTO(00:11) 17,540-18,540 µs and BFO(00:11) 202-302 Hz), end of flight from the run J production recipe (round-3 laws, bank limit U[52, 90] deg, dive branch, C2-silent aim). Inference: waypoints stratum only, 4 × 7×10^5 particles (one tenth of production), 10^4 hand-off rows, 64 descents per row. Core = flight + end of flight, 00:19 option D0 (held out), log-on cause mixture q = 1/3, survival factor. All evidence = core + drift (segment-level, both ocean models, equal weight) + seabed searches (V1 central field at the impact point, no settling). Hydroacoustics and Pléiades excluded. Smoke result, not a calibration result.